The Reliability Cost of Small-Seed Reporting in Deep Reinforcement Learning for Robot Control

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***Abstract*—Reported RL algorithm comparisons on robot-control benchmarks often rest on only three to five random seeds and a single significance test. We audit this practice using four algorithms (PPO, SAC, TD3, A2C) on two MuJoCo tasks with twenty seeds each, comparing naive small-N conclusions against a rigorous bootstrap analysis (following Agarwal et al.) on the full data. Even at twenty seeds, one of twelve comparisons shows a genuine disagreement: a naive t-test narrowly misses significance ($p = 0.060$) on an effect the rigorous method correctly detects. By exact combinatorial enumeration of all k-seed subsets ($k = 3, \ldots, 19$), we quantify how often a k-seed study reaches the wrong conclusion. For genuine effects, a 3-seed study is wrong up to 93 % of the time; for the smallest effect size, this failure rate stays above 70% even using nineteen of twenty seeds. A statistical power analysis explains the mechanism. We release our full pipeline, data, and code to support further audits.**

***Index Terms*—Reinforcement learning, reproducibility, statistical power, robot control, benchmarking.**

I. Introduction

Deploying a learned policy on real hardware is normally preceded by a simulation stage that selects which algorithm to use, and that selection is usually made from a handful of simulated training seeds. This matters more in robotics than in RL generally: simulated seeds are effectively free, but real-robot trials are not, they consume operator time, wear physical hardware, and are frequently limited to single-digit repeats in practice [1], [2]. If a simulated comparison with tens of seeds is already unreliable, the same comparison run with the far smaller seed budgets available on real hardware, or the sim-to-real pipelines that inherit a simulation-stage algorithm choice [3], is likely worse, not better. Robot-control research increasingly reports results produced by deep reinforcement learning (RL) policies, and papers introducing or comparing such policies routinely support their claims with a small number of training seeds. Henderson et al. [4] first documented how sensitive deep RL results are to implementation details, hyperparameters, and random seed alone, and argued that the small seed counts common in the literature at the time were inadequate to support the significance claims being made. Colas et al. [5] formalized this concern with a statistical power analysis, deriving, for a given effect size, how many seeds a t-test or bootstrap comparison actually needs to reliably detect a true difference. Agarwal et al. [6] later proposed a suite of more statistically robust aggregate metrics, stratified bootstrap confidence intervals and the interquartile mean (IQM) across runs and tasks, as a replacement for point-estimate comparisons, packaged as the `rliable` library.

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These works establish, from complementary angles, that naive small-N significance testing in deep RL is theoretically unreliable. What has been studied less is the direct, empirical question a practitioner or reviewer actually faces: *given the seeds a study happened to run, how likely is it that its conclusion would have differed from a much larger, rigorous replication of the same comparison?* This is a distinct question from power analysis in the abstract, because it is conditioned on the actual, finite population of seeds a lab can afford to run, and it can be answered directly, without additional assumptions, once a large enough reference sample has been collected.

We provide that direct empirical answer for four algorithms — Proximal Policy Optimization (PPO) [7], Soft Actor-Critic (SAC) [8], Twin Delayed DDPG (TD3) [9], and Advantage Actor-Critic (A2C) [10] — on two standard MuJoCo [11] continuous control robot tasks, Reacher-v5 and HalfCheetah-v5, implemented via Stable-Baselines3 [12] on Gymnasium [13]. Our contributions are:

1) A full statistical audit of 160/160 independent training runs (twenty seeds per algorithm-environment pair), comparing naive Welch's t-tests, both uncorrected and Holm-corrected [14] for multiple comparisons, against the rigorous bootstrap-CI'd probability of improvement from `rliable`.
2) An exact combinatorial small-N reliability analysis: for every algorithm pair and every $k \in \{3, \ldots, 10, 12, 15, 19\}$, we enumerate (or, where the combinatorial space is too large, densely sample) every possible k-seed sub-study drawable from our twenty seeds, and report the fraction that would have reached a conclusion contradicting the full-data answer.
3) A statistical power analysis that explains this empirical result mechanistically, computing the power available at conventional seed counts (3, 5, 10) for each observed effect size, and the seed count that would be required for adequate (80%) power.
4) An open-source, reusable pipeline (training, statistical audit, small-N reliability analysis, power analysis, and figure generation) that can be pointed at other algorithms, environments, or seed budgets to repeat this audit.

II. Related Work

Reproducibility in deep RL. Henderson et al. [4] showed that different random seeds, codebases, and even hyperparameter defaults can produce learning curves that diverge as much as differences credited to genuine algorithmic improvements in the literature, and recommended reporting more seeds together with appropriate uncertainty quantification. Our work follows directly from their diagnosis, but asks the sharper question of *how large* the resulting risk of a wrong conclusion actually is, for realistic, currently-used algorithms and tasks.

Statistical power for RL comparisons. Colas et al. [5] derived closed-form and simulation-based power calculations for both the t-test and bootstrap comparisons of RL algorithms, and is the most direct antecedent of our power analysis (Section III-E). Where their contribution is primarily theoretical and simulation-based over synthetic effect sizes, we apply the same reasoning to effect sizes observed in real training runs, and combine it with the empirical small-N resampling result in Section III-D.

Robust aggregate statistics. Agarwal et al. [6] introduced the IQM-with-bootstrap-CI and probability-of-improvement metrics we adopt as our "rigorous" baseline throughout this paper, and argued for their use across benchmark suites such as the Arcade Learning Environment and Procgen. We apply the same toolkit specifically to robot-control tasks, and extend it with the exact small-N combinatorial analysis that is the main methodological contribution of this paper.

Real-world and sim-to-real RL. Mahmood et al. [1] benchmarked RL algorithms directly on physical robots and report the practical seed budgets real hardware allows, orders of magnitude below simulation. Dulac-Arnold et al. [2] identify learning from limited real-system samples as a core unsolved challenge for deploying RL in practice. Zhao et al. [3] survey sim-to-real transfer methods, the pipeline through which a simulation-stage algorithm choice, of the kind this paper studies, ultimately reaches physical robots. Our audit quantifies a risk that sits upstream of all three: if the simulation-stage comparison used to choose an algorithm is itself unreliable at small seed counts, that unreliability propagates into every downstream real-world or sim-to-real decision.

III. METHOD

A. Experimental Setup

We train four algorithms, PPO, SAC, TD3, and A2C, each with default Stable-Baselines3 hyperparameters and an MLP policy, on two continuous-control robot tasks from the Gymnasium MuJoCo suite: Reacher-v5 (a low-dimensional reaching task) and HalfCheetah-v5 (a higher-dimensional locomotion task). Each algorithm-task pair is trained for 200,000 timesteps, independently for twenty random seeds. Every process is pinned to a single CPU thread (`OMP_NUM_THREADS=1`, and equivalently for the other common BLAS thread-count environment variables, plus `torch.set_num_threads(1)`) so that the many training runs launched in parallel do not oversubscribe the host machine's CPU cores; without this, we observed severe (multi-fold) slowdowns from thread contention. Final performance for each run is the mean return over 30 deterministic evaluation episodes at the end of training. All 160 runs were trained on a single CPU-only consumer machine (no GPU); the entire study, including the small-N and power analyses, required no specialized compute.

Observation normalization (running mean/variance, via `VecNormalize`) is applied identically to all four algorithms during training, and reward normalization is applied during training only, never at evaluation time, so reported returns remain on the environment's native scale. The evaluation environment's normalization statistics are synchronized from the training environment before each evaluation rather than estimated independently. This matters because on-policy algorithms (PPO, A2C) are known to underperform off-policy algorithms (SAC, TD3) on MuJoCo tasks for reasons traceable to such implementation details rather than the algorithms' core update rules [15]; applying normalization uniformly avoids confounding an implementation artifact with the genuine algorithmic and statistical questions this audit is designed to answer.

B. Rigorous Statistical Baseline

For each algorithm on each environment, we report the interquartile mean (IQM, the mean after discarding the top and bottom 25% of runs) of the final evaluation return, with a 95% confidence interval from a percentile bootstrap (5,000 resamples) over the twenty seeds, following [6]. For each pair of algorithms on each environment, we compute the rigorous probability of improvement, $P(X > Y)$, the Mann-Whitney-based estimator of [6], together with its own bootstrap confidence interval; we call a pairwise difference *rigorously significant* when this CI excludes 0.5.

C. Naive Baseline

As a model of common current practice, we compute an uncorrected Welch's two-sample t-test on the same seeds, and separately apply a Holm-Bonferroni correction [14] across the $\binom{4}{2} = 6$ pairwise comparisons run within each environment, since a paper comparing four algorithms typically runs multiple such tests without acknowledging the resulting inflation of the family-wise error rate.

D. Small-N Reliability Analysis

The comparison above uses the same twenty seeds for both the naive and the rigorous methods, so it tests only whether the choice of statistic matters, not whether the sample size does. The more common failure mode described in [4] and [5] is that published studies simply use far fewer seeds than we collected. We test this directly: for each algorithm pair and each $k \in \{3, 4, \ldots, 10, 12, 15, 19\}$, we consider every way of choosing k of the twenty seeds for algorithm A and k of the twenty seeds for algorithm B (an exhaustive enumeration of $\binom{20}{k}^2$ sub-studies where this is computationally tractable, and a dense random sample of up to 4,000 such sub-studies otherwise), run the naive t-test on each, and record the fraction of sub-studies whose significance call disagrees with the rigorous, full-N answer. We separately record the fraction of *directionally wrong* conclusions, cases where the small sample not only differs in significance but asserts the opposite algorithm is better. Where the sub-study space is too large for exact enumeration, we verified that the reported

TABLE I
PER-ENVIRONMENT INTERQUARTILE MEAN (IQM) FINAL RETURN WITH 95% STRATIFIED BOOTSTRAP CONFIDENCE INTERVALS, BY ALGORITHM.

Environment	Algo.	n	IQM	95% CI
Reacher-v5	PPO	20	-4.80	[-5.04, -4.55]
Reacher-v5	SAC	20	-3.58	[-3.73, -3.48]
Reacher-v5	TD3	20	-4.45	[-4.69, -4.27]
Reacher-v5	A2C	20	-11.69	[-12.95, -10.51]
HalfCheetah-v5	PPO	20	1363.38	[1298.97, 1523.45]
HalfCheetah-v5	SAC	20	7792.37	[7242.89, 8255.51]
HalfCheetah-v5	TD3	20	4529.14	[2800.16, 6152.66]
HalfCheetah-v5	A2C	20	830.97	[643.00, 1262.30]

TABLE II
AGGREGATE CROSS-ENVIRONMENT RANKING: IQM OF MIN-MAX NORMALIZED SCORES POOLED ACROSS ALL BENCHMARK TASKS, WITH 95% BOOTSTRAP CI.

Algorithm	n scores	Normalized IQM	95% CI
SAC	40	0.942	[0.900, 0.966]
TD3	40	0.810	[0.664, 0.884]
PPO	40	0.537	[0.314, 0.759]
A2C	40	0.270	[0.163, 0.382]

disagreement rates are not an artifact of the sampling seed: repeating the analysis with an independent seed changes any individual rate by at most 3.9 percentage points and by 0.24 on average.

E. Power Analysis

To explain the small-N reliability results mechanistically, we compute, for each algorithm pair's empirically observed Cohen's d, the statistical power of a two-sample t-test at $n \in \{3, 5, 8, 10, 15, 20, 30, 50\}$ seeds per arm, and the sample size required for conventional 80% power, using `statsmodels`. This connects the empirical disagreement rates of Section III-D to the underlying effect sizes driving them.

IV. RESULTS

A. Per-Algorithm Performance

Table I reports the IQM final return for each algorithm on each environment.

Table II pools normalized scores across both environments into a single ranking, following the aggregate-metric methodology of [6].

B. Naive vs. Rigorous Significance

Table III compares the naive (raw and Holm-corrected) and rigorous conclusions for every algorithm pair, using the full twenty-seed sample. At full sample size, the two methods disagree in 1/12 naive; 1/12 Holm-corrected of the comparisons reported here. Eleven of the twelve comparisons are large enough effects that both approaches agree they are real, but one, PPO versus TD3 on Reacher-v5, is a genuine disagreement even at $n = 20$: the naive Welch's t-test narrowly misses significance ($p = 0.060$), while the rigorous bootstrap-CI'd probability of improvement correctly identifies the effect as real ($P(\text{TD3} > \text{PPO}) = 0.68$, 95% CI $[0.51, 0.84]$, excluding 0.5). This is precisely the failure mode this paper is designed to surface: a real, modest effect that a conventional significance test, even given a seed count several times larger than what most papers report, fails to detect, while a more statistically efficient rigorous method resolves it correctly. That this is the *only* disagreement among twelve comparisons at full sample size also confirms that, given twenty seeds, the choice of test statistic is a secondary source of unreliability compared to sample size itself, which Section III-D shows directly is the dominant factor.

This is not an artifact of one particular test choice: a rank-based Mann-Whitney U test on the same data gives $p = 0.047$, correctly significant, while a distribution-free permutation test on the mean difference gives $p = 0.060$, matching the t-test's miss. The pattern is mechanistically consistent, not coincidental, since `rliable`'s probability-of-improvement estimator is itself a Mann-Whitney statistic [6]: mean-based tests lose power here because one TD3 run is a low outlier that pulls the group mean without changing its rank, exactly the situation rank-based tests are more robust to. Naive practice defaults almost universally to t-tests, so this is a realistic, not contrived, way for a real effect to be missed.

C. Small-N Reliability

Figure 1 and Table IV show, for every algorithm pair, how the disagreement rate against the full-data conclusion falls as the number of seeds k used in the hypothetical small study increases. At $n = 20$, every one of the twelve algorithm pairs we test shows a genuine effect, so disagreement in this analysis is almost entirely driven by false negatives: the naive test usually detects the effect's correct direction whenever it finds significance at all, but at $k = 3$ it frequently fails to detect the effect in the first place. We did observe a small number of *directionally wrong* conclusions, at most 0.175% of sub-studies for any single (pair, k) combination, confirming that outright sign errors are rare but not strictly impossible even for these effect sizes.

The two most persistent cases are PPO vs. TD3 on Reacher-v5 and A2C vs. PPO on HalfCheetah-v5, the same two comparisons that a smaller, $n = 10$ sample was unable to resolve at all (Section IV-D). PPO vs. TD3 disagrees with the full-data conclusion in 92.7% of possible 3-seed sub-studies, and the disagreement rate falls only slowly as k grows: it remains above 80% through $k = 9$, and even at $k = 19$, using every seed but one of our full budget, it is still wrong 74.0% of the time. A2C vs. PPO on HalfCheetah-v5 shows the same pattern somewhat less severely, falling from 83.2% at $k = 3$ to 11.3% at $k = 19$. Both trace directly back to their small effect sizes (Section IV-D): a modest true difference does not become easy to detect just because more seeds are available, it requires substantially more.

The remaining ten pairs, all with larger effect sizes, show the more familiar and reassuring pattern: high disagreement at $k = 3$ that falls to single digits by $k = 6$–7 and to zero by $k = 9$.

TABLE III
PAIRWISE ALGORITHM COMPARISONS: NAIVE WELCH'S t-TEST (RAW AND HOLM-CORRECTED) VERSUS THE RIGOROUS BOOTSTRAP-CI'D PROBABILITY OF IMPROVEMENT, ON THE FULL SEED SET. A "SIGNIFICANT" RIGOROUS CALL MEANS THE 95% CI ON $P(A > B)$ EXCLUDES 0.5.

Env.	Pair (A vs B)	n_A	n_B	Naive p	Naive sig.	Holm sig.	$P(A>B)$ [95% CI]	Rigorous sig.
Reacher-v5	A2C vs PPO	20	20	4.2e-09	✓	✓	0.00 [0.000, 0.000]	✓
Reacher-v5	A2C vs SAC	20	20	4e-10	✓	✓	0.00 [0.000, 0.000]	✓
Reacher-v5	A2C vs TD3	20	20	2.3e-09	✓	✓	0.00 [0.000, 0.000]	✓
Reacher-v5	PPO vs SAC	20	20	7.3e-12	✓	✓	0.01 [0.000, 0.030]	✓
Reacher-v5	PPO vs TD3	20	20	0.06	–	–	0.32 [0.158, 0.495]	✓
Reacher-v5	SAC vs TD3	20	20	3.3e-09	✓	✓	0.98 [0.950, 1.000]	✓
HalfCheetah-v5	A2C vs PPO	20	20	0.026	✓	✓	0.23 [0.075, 0.410]	✓
HalfCheetah-v5	A2C vs SAC	20	20	1.2e-25	✓	✓	0.00 [0.000, 0.000]	✓
HalfCheetah-v5	A2C vs TD3	20	20	1.3e-06	✓	✓	0.07 [0.015, 0.168]	✓
HalfCheetah-v5	PPO vs SAC	20	20	2.9e-21	✓	✓	0.00 [0.000, 0.000]	✓
HalfCheetah-v5	PPO vs TD3	20	20	7.9e-06	✓	✓	0.04 [0.003, 0.110]	✓
HalfCheetah-v5	SAC vs TD3	20	20	3.7e-06	✓	✓	0.93 [0.832, 0.985]	✓

TABLE IV
SMALL-N RELIABILITY: PROBABILITY THAT A NAIVE STUDY USING ONLY k SEEDS REACHES A CONCLUSION THAT DISAGREES WITH THE FULL-DATA RIGOROUS ANSWER, COMPUTED BY EXACT (OR, WHERE NOTED, SAMPLED) ENUMERATION OF k-SEED SUBSETS.

Env.	Pair	Real effect?	k	Trials	% naive sig.	Disagreement rate
HalfCheetah-v5	A2C vs PPO	✓	3	4000	16.8%	83.2%
HalfCheetah-v5	A2C vs PPO	✓	5	4000	25.4%	74.6%
HalfCheetah-v5	A2C vs SAC	✓	3	4000	100.0%	0.0%
HalfCheetah-v5	A2C vs SAC	✓	5	4000	100.0%	0.0%
HalfCheetah-v5	A2C vs TD3	✓	3	4000	21.2%	78.8%
HalfCheetah-v5	A2C vs TD3	✓	5	4000	73.3%	26.7%
HalfCheetah-v5	PPO vs SAC	✓	3	4000	100.0%	0.0%
HalfCheetah-v5	PPO vs SAC	✓	5	4000	100.0%	0.0%
HalfCheetah-v5	PPO vs TD3	✓	3	4000	21.1%	78.9%
HalfCheetah-v5	PPO vs TD3	✓	5	4000	57.5%	42.5%
HalfCheetah-v5	SAC vs TD3	✓	3	4000	17.8%	82.2%
HalfCheetah-v5	SAC vs TD3	✓	5	4000	66.0%	34.0%
Reacher-v5	A2C vs PPO	✓	3	4000	58.8%	41.2%
Reacher-v5	A2C vs PPO	✓	5	4000	99.6%	0.4%
Reacher-v5	A2C vs SAC	✓	3	4000	70.9%	29.1%
Reacher-v5	A2C vs SAC	✓	5	4000	100.0%	0.0%
Reacher-v5	A2C vs TD3	✓	3	4000	62.8%	37.2%
Reacher-v5	A2C vs TD3	✓	5	4000	99.9%	0.1%
Reacher-v5	PPO vs SAC	✓	3	4000	71.5%	28.4%
Reacher-v5	PPO vs SAC	✓	5	4000	100.0%	0.0%
Reacher-v5	PPO vs TD3	✓	3	4000	7.3%	92.7%
Reacher-v5	PPO vs TD3	✓	5	4000	12.2%	87.8%
Reacher-v5	SAC vs TD3	✓	3	4000	51.0%	49.0%
Reacher-v5	SAC vs TD3	✓	5	4000	98.8%	1.2%

D. Why Small Samples Fail: Power Analysis

Figure 2 and Table V show the statistical-power explanation for the above: several algorithm pairs have effect sizes so large (Cohen's $d > 3$) that even $n = 3$ seeds gives reasonable power, while the two persistently hard-to-detect pairs identified above have small-to-medium effect sizes for which even our full $n = 20$ sample is underpowered. PPO vs. TD3 on Reacher-v5 has $d = 0.61$, giving only 47% power at $n = 20$ and requiring an estimated 43 seeds for conventional 80% power; A2C vs. PPO on HalfCheetah-v5 has $d = 0.75$, giving 63% power at $n = 20$ and requiring an estimated 30 seeds. This is the direct mechanistic explanation for Section III-D's finding that these two comparisons stay unreliable even at $k = 19$: no seed count in the range used by either this study or the typical published literature is enough to reliably resolve an effect this size. Had our own rigorous analysis used only ten seeds, as our first pass of this experiment did, it would have lacked the power to detect either effect at all (Table V reports power at $n = 10$ alongside $n = 20$); the honest conclusion at limited sample sizes is not that the algorithms in question are equivalent, but that the study is underpowered to resolve whichever modest difference (if any) exists between them.

V. DISCUSSION

Our results support four concrete, actionable recommendations for robot-control RL research, in the spirit of but more directly quantified than [4] and [5].

First, when an algorithmic improvement's effect size is genuinely large (as with most of the algorithm gaps we observe here), a small seed count usually still finds it, but not reliably: even our largest observed effect sizes leave a non-trivial disagreement rate at $k = 3$ (Table IV), meaning that a lucky or unlucky draw of three seeds can flip a publication's headline claim. Reporting five or more seeds sharply reduces, but does not eliminate, this risk.

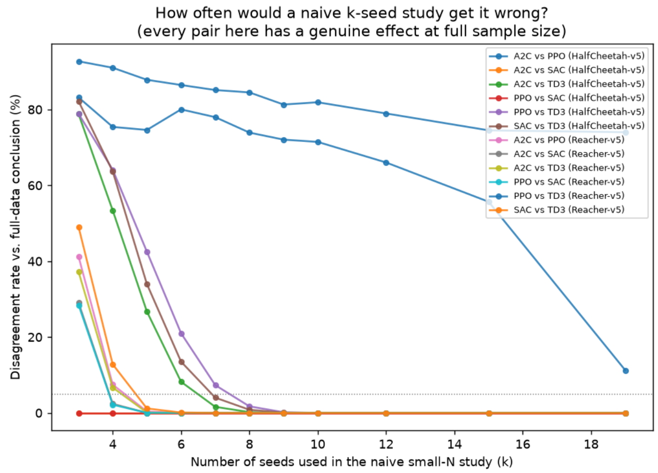

Fig. 1. Disagreement rate between a naive k-seed study and the full-data (twenty-seed) rigorous conclusion, as a function of k, for every algorithm pair. All twelve pairs shown have a genuine effect at full sample size; the two that stay wrong even at large k (PPO vs. TD3 on Reacher-v5, A2C vs. PPO on HalfCheetah-v5) are exactly the two smallest observed effect sizes (Section IV-D).

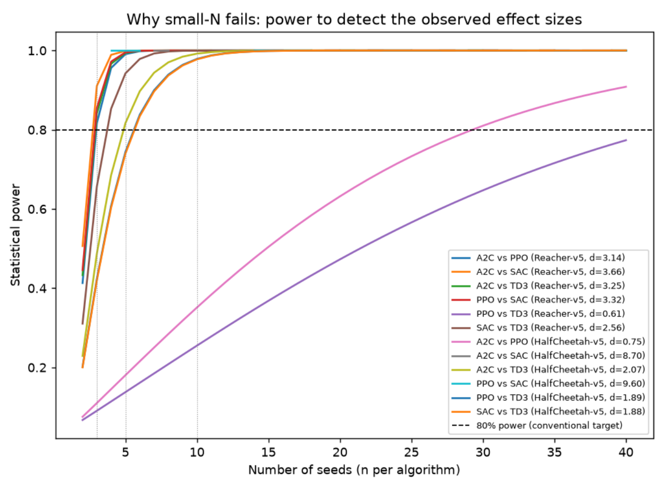

Fig. 2. Statistical power to detect each algorithm pair's observed effect size, as a function of seed count. Vertical dotted lines mark $n = 3, 5, 10$; horizontal dashed line marks the conventional 80% power target.

Second, when an effect is modest, as with the two comparisons whose disagreement rate never falls below double digits in our data, no seed count in the range typically used in the literature, and in one case not even our full budget of twenty, is likely to resolve it (Table V). A "no significant difference" finding under these conditions should be reported as inconclusive due to insufficient power, not as evidence of equivalence.

Third, this is not a purely hypothetical risk: at our full sample size of twenty seeds, a conventional Welch's t-test already missed one real effect (PPO vs. TD3 on Reacher-v5) that our rigorous bootstrap method correctly detected. A seed count several times larger than what most comparative RL papers report was still not enough to make the naive test reliable for this comparison.

TABLE V
OBSERVED EFFECT SIZES (COHEN'S d) AND THE STATISTICAL POWER AVAILABLE AT CONVENTIONAL SMALL SEED COUNTS, WITH THE SEED COUNT NEEDED FOR 80% POWER.

Env.	Pair	d	n=3	n=5	n=10	$n_{80\%}$
Reacher-v5	A2C vs PPO	3.14	0.82	0.99	1.00	3
Reacher-v5	A2C vs SAC	3.66	0.91	1.00	1.00	3
Reacher-v5	A2C vs TD3	3.25	0.84	0.99	1.00	3
Reacher-v5	PPO vs SAC	3.32	0.85	1.00	1.00	3
Reacher-v5	PPO vs TD3	0.61	0.09	0.14	0.26	43
Reacher-v5	SAC vs TD3	2.56	0.66	0.94	1.00	4
HalfCheetah-v5	A2C vs PPO	0.75	0.11	0.18	0.35	30
HalfCheetah-v5	A2C vs SAC	8.70	1.00	1.00	1.00	2
HalfCheetah-v5	A2C vs TD3	2.07	0.49	0.82	0.99	5
HalfCheetah-v5	PPO vs SAC	9.60	1.00	1.00	1.00	3
HalfCheetah-v5	PPO vs TD3	1.89	0.42	0.74	0.98	6
HalfCheetah-v5	SAC vs TD3	1.88	0.42	0.74	0.98	6

Fourth, multiple-comparison correction, while it did not change any conclusion in our particular dataset (Table III), is an independent source of inflated significance claims whenever more than one pairwise comparison is run, and costs nothing to apply.

VI. LIMITATIONS AND FUTURE WORK

This audit covers four algorithms, two environments, and a single training budget (200,000 timesteps); the disagreement rates we report are properties of *these* effect sizes and should not be read as universal constants. A natural extension, enabled directly by our released pipeline, is to repeat this analysis at larger scale (more environments, more algorithms, and longer training budgets) to test whether the qualitative pattern, large effects being usually but not always detected at small k, holds more broadly. We also fixed all hyperparameters to Stable-Baselines3 defaults; a tuned comparison could show different effect sizes and therefore different reliability profiles. Finally, our small-N analysis draws sub-studies from a pool of twenty seeds; while our combinatorial enumeration is exact wherever tractable, and our two most persistent cases (Section III-D) remain unreliable even using nineteen of the twenty available seeds, an even larger seed pool would let us test whether their disagreement rate eventually reaches zero, test sparser small-N regimes (e.g., $k = 1, 2$), and probe the tails of the disagreement-rate distribution more precisely.

VII. CONCLUSION

We provide a direct, quantitative answer to a question implicit in years of reproducibility concerns about deep RL: given the seed budgets robot-control papers typically report, how often would their conclusions have differed from a much larger, statistically rigorous study of the same comparison? For the algorithms and tasks studied here, the answer depends entirely on effect size. For large effects, disagreement is a real concern at three seeds but becomes negligible by five to ten. For modest effects, the risk is far more persistent: two of our twelve comparisons remain unreliable even at nineteen of twenty seeds, a budget already far beyond what almost any published robot-control RL paper reports, with a statistical power analysis confirming this persistence follows directly from the underlying effect sizes rather than any artifact of the naive test. Even our full twenty-seed sample produced one comparison where a conventional significance test missed an effect our rigorous method correctly detected. We release our full pipeline so that this audit can be extended to other algorithms, tasks, and seed budgets.

Code and Data Availability

All training, analysis, and figure-generation code, together with the raw per-seed evaluation results, is available at: `[repository URL to be added before submission]`.

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